LungLogic: Model-Based Airflow Control with Measured-Delay Compensation and Auto PID

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1. Introduction

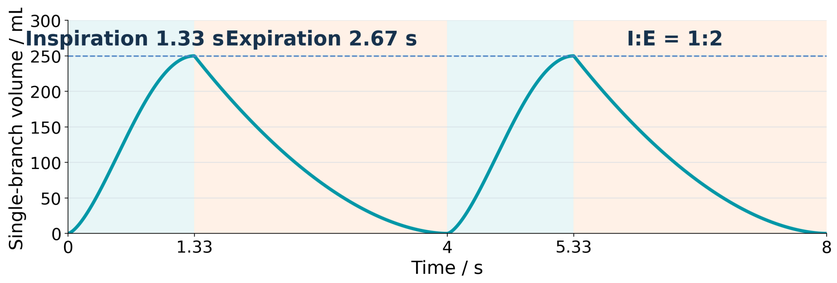
1.1 Project need

Dynamic lung phantoms support imaging and ventilation research. Existing products and laboratory systems do not combine programmable timing, repeatable volume and independent left/right actuation.

1.2 Project objective

LungLogic develops a pneumatic and software platform to control a medical biomimetic lung model. This stage validates one 250 mL branch before bilateral integration.

1.3 Nominal breathing reference



Breathing-cycle illustration at 15 bpm and I:E = 1:2.

RR · respiratory rate	12–16 bpm; nominal 15 bpm
I:E · inhale:exhale time	1:2; 1.33 s / 2.67 s at 15 bpm
VT · tidal volume	250 mL per branch
Repeatability	VT error within ±5%

Nominal design values, not measured performance. The 250 mL branch target is half of the ~500 mL adult tidal-volume example.

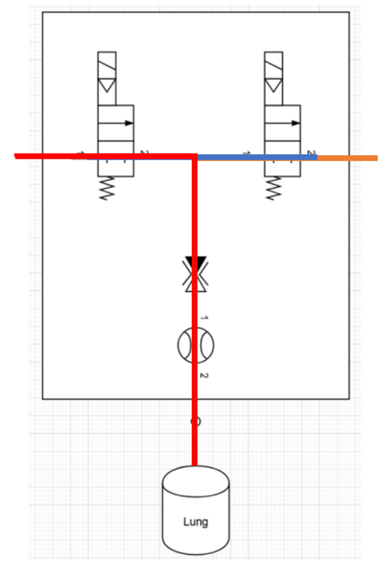
Source: 4DMedical project scope, 19 March 2026, pp. 5 and 13. The ±5% repeatability value is a design target.

2. Pneumatic System & Hardware

2.1 Objective

Control one lung branch and measure airflow for closed-loop regulation and calibration.

2.2 Pneumatic system prototype



Prototype photograph (to be added)

Fig. 1. Single-lung route; prototype photograph pending.

2.3 Embedded control architecture

PC / Python UI → **Controller** → **24 V bridge PCB** → **Valves + sensor**

The controller runs the 100 Hz breathing and flow loop. The bridge connects logic to the routing valves, PPV and SFM3300 sensor.



Fig. 2. Control architecture and bridge PCB.

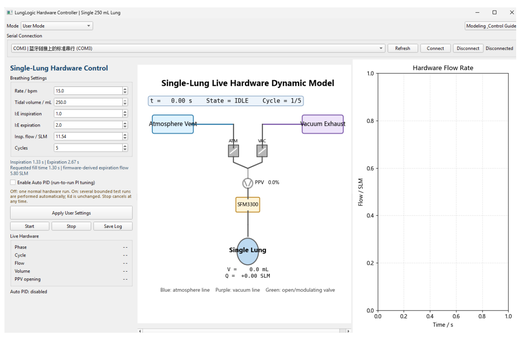
3. Integrated Control System & Software

3.1 Objective

Track flow and tidal volume with a visible, repeatable closed-loop system. Feedforward supplies the nominal valve command; sensor feedback corrects error.

3.2 Operator and developer interface

User Mode



Developer Mode

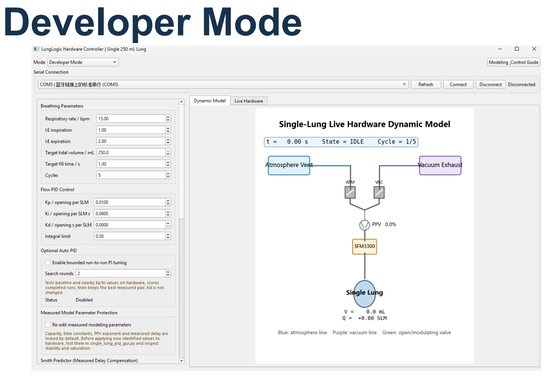
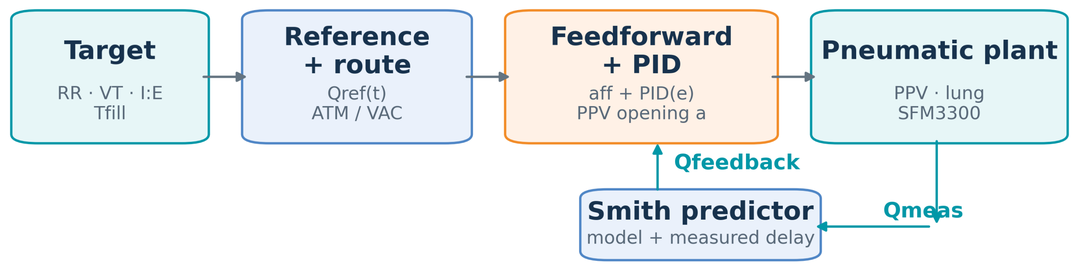


Fig. 3. User and Developer modes.

3.3 Model-assisted flow feedback



$$Q_{ref,\,in} = 0.06 V_T / T_{fill} \quad \cdot \quad e = Q_{ref} - |Q_{feedback}| \quad \cdot \quad a = \mathrm{clip}(a_{ff} + PID(e), 0, 1)$$

Fig. 4. Feedback control with optional Smith prediction.

3.4 Auto PID and measured-delay compensation

Auto PID scores bounded Kp/Ki trials. Smith prediction uses the identified model and measured delay; hardware adds no artificial delay.

4. Results & Future Work

4.1 Experimental results

EXPERIMENTAL RESULTS TO BE ADDED

Insert measured flow and volume plots
Report: VT error · flow RMSE · repeatability

Fig. 5. Reserved for measured hardware results.

4.2 Future work

1. **Integrate the biomimetic lung model**
2. **Re-identify the model and delay**
3. **Validate bilateral repeatability**

Extend the verified branch to independent left/right control across the required breathing range.